Exact SU(2) Wilson transfer operator and the finite-spatial-lattice time limit

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1 Retained action, coefficients, and conventions

Fix $a > 0$ and $g_{\mathrm{YM}} > 0$. The spatial regulator is a fixed finite oriented cubical graph with vertices V, physical spatial edges E, and the full set P_s of its geometric spatial plaquettes, each included once, with $|\mathsf{E}| \geq 1$. Open spatial boundary conditions or a fixed periodic spatial lattice are both allowed. Reverse traversal uses U_e^{-1}, not a second independent link. Write $N = |\mathsf{E}|$ and

$$\mathcal{Q} = \mathrm{SU}(2)^{\mathsf{E}}, \qquad \mathcal{H} = L^2(\mathcal{Q}, \lambda), \qquad d\lambda = \prod_{e \in \mathsf{E}} dU_e,$$

with probability Haar on each link. The spatial graph, a, and g_{YM} stay fixed in every limit in this paper.

The original invariant Lie metric and continuum action are

$$c(X, Y) = -\frac{1}{2}\operatorname{tr}(XY), \qquad S(\mathscr{A}) = \frac{1}{2g_{\mathrm{YM}}^2} \int \sum_{\mu<\nu} c(F_{\mu\nu}, F_{\mu\nu})\, d^4x. \tag{1}$$

For time spacing $\epsilon > 0$, retain exactly

$$b_t(\epsilon) = \frac{a}{2g_{\mathrm{YM}}^2 \epsilon}, \qquad b_s(\epsilon) = \frac{\epsilon}{2g_{\mathrm{YM}}^2 a}. \tag{2}$$

For the frozen-curvature holonomy $e^{\delta F}$, $2 - \operatorname{Re}\operatorname{tr} e^{\delta F} = \delta^2 c(F, F) + O_F(\delta^4)$ on $\mathfrak{su}(2)$, by the exponential series, tracelessness and the paired eigenvalues of F. The exact coefficient checks are

$$\frac{b_t(a\epsilon)^2}{a^3\epsilon} = \frac{1}{2g_{\mathrm{YM}}^2}, \qquad \frac{b_s a^4}{a^3\epsilon} = \frac{1}{2g_{\mathrm{YM}}^2}.$$

These equalities record the prescribed classical action convention. The transfer calculation below uses the full compact plaquette action, not a curvature truncation, and proves no spatial continuum limit.

Choose $T_1, T_2, T_3 \in \mathfrak{su}(2)$ with $-2\operatorname{tr}(T_a T_b) = \delta_{ab}$, for example $T_a = i\sigma_a/2$ with the Pauli matrices. Define

$$X_{e,a} f(U) = \left.\frac{d}{dt}\right|_0 f(\ldots, e^{tT_a} U_e, \ldots), \qquad \Delta_e^T = \sum_{a=1}^{3} X_{e,a}^2.$$

In the original metric, $c(T_a, T_b) = \delta_{ab}/4$, so the c-orthonormal basis is $S_a = 2T_a$, and

$$X_{e,a}^c = 2X_{e,a}, \qquad \Delta_e^c = 4\Delta_e^T. \tag{3}$$

The probability Haar convention changes no operator coefficient. If $d\,\mathrm{vol}_c = \mathcal{V}_c d\lambda$ on this compact product, $f \mapsto \sqrt{\mathcal{V}_c} f$ is the exact unitary change from $L^2(d\,\mathrm{vol}_c)$ to $L^2(d\lambda)$ and intertwines the same differential operators.

The original inverse-transport gauge law and the convention used in the companion Hamiltonian derivation are related at the configuration level by

$$(\beta_h U)_e = h_{s(e)}^{-1} U_e h_{t(e)}, \qquad (\alpha_g U)_e = g_{s(e)} U_e g_{t(e)}^{-1}, \qquad \alpha_g U = \beta_h U \text{ when } g_v = h_v^{-1} \text{ for every } v. \tag{4}$$

Inversion preserves Haar, so their vertex gauge averages are the same operator. No link variable is inverted by this dictionary; only the parameter describing the same vertex change is replaced by its inverse.

2 The exact single-link character multiplier

For $b > 0$, let

$$Z(b) = \int_{\mathrm{SU}(2)} e^{b\,\mathrm{Re}\,\mathrm{tr}\,U} dU, \qquad k_b(U) = \frac{e^{b\,\mathrm{Re}\,\mathrm{tr}\,U}}{Z(b)}, \qquad (\mathcal{K}_b f)(U') = \int k_b(U'U^{-1}) f(U) dU. \tag{5}$$

Every $U \in \mathrm{SU}(2)$ has eigenvalues $e^{i\theta}, e^{-i\theta}$ with $0 \le \theta \le \pi$. Probability Haar integration of a class function is

$$\int f(U) dU = \frac{2}{\pi} \int_0^\pi f(\theta) \sin^2\theta \, d\theta.$$

For example, write $U = w_0 I + i \sum_{a=1}^3 w_a \sigma_a$ on the unit three-sphere. Slicing at $w_0 = \cos\theta$ gives density $(2/\pi)\sqrt{1 - w_0^2} dw_0$, which is the displayed formula.

Define the integer-order modified Bessel functions by their entire series

$$I_d(2b) = \sum_{r=0}^\infty \frac{b^{2r+d}}{r!(r+d)!}, \qquad d \in \mathbb{Z}_{\ge 0}. \tag{6}$$

They also satisfy

$$I_d(2b) = \frac{1}{\pi} \int_0^\pi e^{2b\cos\theta} \cos(d\theta)\, d\theta. \tag{7}$$

To prove this, multiply the uniformly absolutely convergent series for e^{bz} and $e^{b/z}$ on $|z| = 1$. The coefficient of z^d is (6), and Fourier orthogonality gives (7). For $d \ge 1$, integration of the derivative of $e^{2b\cos\theta}\sin(d\theta)$ over $[0, \pi]$ gives

$$I_{d-1}(2b) - I_{d+1}(2b) = \frac{d}{b} I_d(2b). \tag{8}$$

The endpoint term is zero, and $2\sin\theta\sin(d\theta) = \cos((d-1)\theta) - \cos((d+1)\theta)$ fixes every coefficient in this recurrence.

The spin-j irreducible representation has $j \in \frac{1}{2}\mathbb{Z}_{\ge 0}$, dimension $d = 2j + 1$, and character

$$\chi_j(\theta) = \sum_{r=0}^{d-1} e^{i(d-1-2r)\theta} = \frac{\sin(d\theta)}{\sin\theta}.$$

The finite sum is the character of the $(d-1)$st symmetric power of the defining representation; its equality with the sine quotient is the finite geometric-series identity, including endpoint values by continuity. These are all irreducibles of SU(2), and their matrix coefficients are complete by Peter–Weyl [1].

Theorem 1. *The exact normalization and the exact spin-j multiplier of* (5) *are*

$$Z(b) = \frac{I_1(2b)}{b}, \qquad \mathcal{K}_b D^j_{rs} = m_j(b) D^j_{rs}, \qquad m_j(b) = \frac{I_{2j+1}(2b)}{I_1(2b)}. \tag{9}$$

For every j and $b > 0$, $0 < m_j(b) \le 1$, and $m_0(b) = 1$. The operator $\mathcal{K}_b$ is a positive self-adjoint contraction, is injective, and is trace class.

Proof. The exact unnormalized character integral is

$$\begin{aligned}\int e^{b\,\mathrm{Re\,tr}\,U} \chi_j(U) dU &= \frac{2}{\pi} \int_0^\pi e^{2b\cos\theta} \sin\theta \sin(d\theta) d\theta \\ &= I_{d-1}(2b) - I_{d+1}(2b) = \frac{d}{b} I_d(2b).\end{aligned}$$

Set $d = 1$ to obtain $Z(b)$. Since k_b is central, the matrix $\int k_b(W) D^j(W^{-1}) dW$ commutes with the representation and hence is scalar. Its trace divided by d is $(dZ(b))^{-1} \int e^{b\,\mathrm{Re\,tr}\,W} \chi_j(W) dW = I_d/I_1$. The substitution $U = W^{-1}U'$ in (5) proves the asserted matrix-coefficient eigenvalue. In particular the dimension factor in the character integral is exactly canceled by division by d; no dimension factor remains in the multiplier.

All terms in (6) are positive for $b > 0$, which proves $m_j > 0$. The integral expression for the normalized multiplier and $|\chi_j(U)| \le d$ prove $|m_j| \le 1$. The matrix coefficients form an orthogonal complete basis, so these real positive eigenvalues prove positivity, self-adjointness, contraction, and injectivity. For fixed b, (6) gives $I_d(2b) \le b^d e^{b^2}/d!$, since $(r+d)! \ge d!$. Thus

$$\mathrm{tr}\,\mathcal{K}_b = \sum_{d=1}^{\infty} d^2 \frac{I_d(2b)}{I_1(2b)} < \infty,$$

where the multiplicity is exactly d^2. The factorial majorant proves the convergence and hence trace-classness. □

3 Controlled small-time derivative of the full compact kernel

Lemma 2. *For each fixed spin j,*

$$m_j(b) = 1 - \frac{j(j+1)}{b} + O_j(b^{-2}) \qquad (b \longrightarrow \infty). \tag{10}$$

The estimate is proved from the exact compact integral and does not replace that integral by a different finite-b kernel.

Proof. Set $y = 1 - \cos\theta \in [0, 2]$. Under the probability density $k_b dU$, the density of y is proportional to $e^{-2by} y^{1/2} (2-y)^{1/2} dy$. Write

$$J_q(b) = \int_0^2 e^{-2by} y^{q+1/2} (2-y)^{1/2} dy, \qquad q = 0, 1, 2.$$

On $[0, 1]$, Taylor's theorem with a bounded second derivative gives $\sqrt{2-y} = \sqrt{2}(1 - y/4) + R(y)$, with $|R(y)| \le Cy^2$. The part on $[1, 2]$ is bounded by a constant times e^{-2b}. The tails when extending the two leading integrals to $[0, \infty)$ have the same exponentially small order times a

polynomial in b. Finally, the error integral over $[0,1]$ is bounded by $C\int_0^\infty e^{-2by}y^{q+5/2}dy$. Using the exact substitution $z=2by$ in these gamma integrals gives

$$J_q(b)=\frac{\sqrt{2}\,\Gamma(q+3/2)}{(2b)^{q+3/2}}\left[1-\frac{q+3/2}{8b}+O_q(b^{-2})\right]. \tag{11}$$

Every error assertion follows from the displayed remainder bound and the exponentially small tail; none extends a finite-b probability measure by declaration. Since $J_0>0$, division gives

$$\mathbb{E}_b[y]=\frac{J_1}{J_0}=\frac{3}{4b}+O(b^{-2}),\qquad \mathbb{E}_b[y^2]=\frac{J_2}{J_0}=\frac{15}{16b^2}+O(b^{-3}).$$

The character χ_j is the polynomial $U_{d-1}(\cos\theta)$, where $U_0(x)=1$, $U_1(x)=2x$, and $U_{r+1}(x)=2xU_r(x)-U_{r-1}(x)$. Induction at $x=1$ gives

$$U_{d-1}(1)=d,\qquad U'_{d-1}(1)=\frac{d(d^2-1)}{3}.$$

For the derivative formula, differentiate the recurrence; the induction step is $U'_{r+1}(1)=2(r+1)+2U'_r(1)-U'_{r-1}(1)$, which the displayed cubic expression satisfies with its initial values. Polynomial Taylor expansion on the whole compact interval $y\in[0,2]$ therefore yields

$$\frac{\chi_j(\theta)}{d}=1-\frac{d^2-1}{3}y+R_j(y),\qquad |R_j(y)|\le C_jy^2.$$

Its exact $\mathbb{E}_b$ average is $m_j(b)$. Substitute the two moment estimates and use $(d^2-1)/4=j(j+1)$ to prove (10). □

Substitution of the retained b_t gives the exact infinitesimal coefficient

$$m_j(b_t(\epsilon))=1-\epsilon\frac{2g_{\mathrm{YM}}^2}{a}j(j+1)+O_j(\epsilon^2). \tag{12}$$

For the chosen basis, $-\Delta^T$ has eigenvalue $j(j+1)$ on spin j. Indeed take the displayed Pauli basis and put $J_a=-i\,dD^j(T_a)$. Its anti-Hermitian generators satisfy $[T_a,T_b]=-\varepsilon_{abc}T_c$, so the Hermitian generators satisfy $[J_a,J_b]=i\varepsilon_{abc}J_c$. Let $J_\pm=J_1\pm iJ_2$. Directly from these commutators, $[J_3,J_\pm]=\pm J_\pm$ and $\sum_a J_a^2=J_-J_++J_3^2+J_3$. On the highest vector $e_1^{\otimes 2j}$ of the symmetric power, J_3 has eigenvalue j and J_+ is zero, so the sum of squares has value $j(j+1)$. Expanding its commutator with each J_a shows it commutes with all three; irreducibility therefore extends this value to the whole representation. An orthogonal change to any other specified orthonormal T basis leaves the sum of squares unchanged. For $j=1/2$, it is explicitly $-\sum_a T_a^2=3I/4$, fixing the absolute scale. Thus (12) has precisely the Δ^T convention, and (3) supplies the other metric.

On the full finite edge product define

$$(K_\epsilon f)(U')=\int_{\mathcal{Q}}\prod_{e\in\mathsf{E}}k_{b_t(\epsilon)}(U'_eU_e^{-1})f(U)d\lambda(U),\qquad H_0=-\frac{2g_{\mathrm{YM}}^2}{a}\sum_e\Delta_e^T. \tag{13}$$

A product matrix coefficient with spins $\boldsymbol{j}=(j_e)$ has K_ϵ eigenvalue $\prod_e m_{j_e}(b_t(\epsilon))$ and H_0 eigenvalue $(2g_{\mathrm{YM}}^2/a)\sum_e j_e(j_e+1)$. For every finite linear combination f of such coefficients,

$$\lim_{\epsilon\downarrow 0}\frac{K_\epsilon f-f}{\epsilon}=-H_0f. \tag{14}$$

This follows by the finite product expansion of (12), not by an assertion uniform over arbitrary spins. Contraction and density then imply $K_\epsilon f\to f$ for every $f\in\mathcal{H}$: approximate f by a fixed finite combination and bound the two approximation errors by its distance from f.

4 Exact magnetic factor and temporal-link Gauss integration

For the ordered spatial plaquette holonomy U_p, set

$$V(U) = \frac{1}{2g_{\mathrm{YM}}^2 a} \sum_{p \in \mathsf{P}_s} (2 - \operatorname{Re}\operatorname{tr} U_p), \qquad M_\epsilon f(U) = e^{-\epsilon V(U)/2} f(U). \tag{15}$$

Every magnetic plaquette is present in this sum. It is a bounded nonnegative real smooth multiplication operator, with $0 \le V \le 2|\mathsf{P}_s|/(g_{\mathrm{YM}}^2 a)$. In particular $\epsilon V = b_s \sum_p (2 - \operatorname{Re}\operatorname{tr} U_p)$ exactly.

Define the vertex gauge average

$$(P_G f)(U) = \int_{\mathrm{SU}(2)^{\mathsf{V}}} f(\alpha_g U) \prod_v dg_v. \tag{16}$$

Every pullback in the integrand is unitary. Haar inversion proves $P_G^* = P_G$, Haar invariance in two successive integrals proves $P_G^2 = P_G$, and another left translation of the integration variable shows that the range consists exactly of the invariant functions. Thus P_G is the orthogonal projection onto $\mathcal{H}_{\mathrm{inv}}$. Equation (4) proves that it is also the original-h Gauss average. Gauge invariance of spatial Wilson traces proves $M_\epsilon P_G = P_G M_\epsilon$. The identity

$$(\alpha_g U')_e (\alpha_g U)_e^{-1} = g_{s(e)} U'_e U_e^{-1} g_{s(e)}^{-1}$$

and centrality of k_b prove that K_ϵ commutes with P_G.

To compute the temporal integral, write U and U' for two adjacent spatial slices and q_v for the temporal link at vertex v from the first slice to the second. In the original inverse-transport convention, the oriented temporal plaquette associated to edge $e : s \to t$ is

$$P_e(U', q, U) = U_e q_t (U'_e)^{-1} q_s^{-1}. \tag{17}$$

Its inverse is $q_s U'_e q_t^{-1} U_e^{-1}$, and $\operatorname{Re}\operatorname{tr} A^{-1} = \operatorname{Re}\operatorname{tr} A$ for a unitary matrix. Hence

$$\operatorname{Re}\operatorname{tr} P_e = \operatorname{Re}\operatorname{tr}[(\alpha_q U')_e U_e^{-1}]. \tag{18}$$

This equality includes every temporal vertex variable; it makes no global temporal gauge-fixing assumption. Under independent original vertex changes $h^{(0)}$ and $h^{(1)}$ on the two slices,

$$U_e \mapsto (h_s^{(0)})^{-1} U_e h_t^{(0)}, \quad U'_e \mapsto (h_s^{(1)})^{-1} U'_e h_t^{(1)}, \quad q_v \mapsto (h_v^{(0)})^{-1} q_v h_v^{(1)}.$$

Multiplication in (17) cancels the intermediate factors and gives $P_e \mapsto (h_s^{(0)})^{-1} P_e h_s^{(0)}$. This verifies directly its original gauge law.

Define the raw two-slice Wilson kernel, with symmetric magnetic endpoint weights, by

$$\mathcal{T}_\epsilon^{\mathrm{raw}}(U', U) = e^{-\epsilon V(U')/2} e^{-\epsilon V(U)/2} \int \exp\left[-b_t \sum_e (2 - \operatorname{Re}\operatorname{tr} P_e(U', q, U)) \right] \prod_v dq_v. \tag{19}$$

The integral is over the whole non-Abelian group at every temporal vertex. With

$$c_\epsilon = [e^{-2b_t} Z(b_t)]^N, \qquad F_\epsilon = M_\epsilon K_\epsilon M_\epsilon, \tag{20}$$

equations (5) and (18) give the exact operator identity

$$\mathcal{T}_\epsilon^{\mathrm{raw}} = c_\epsilon M_\epsilon P_G K_\epsilon M_\epsilon = c_\epsilon F_\epsilon P_G = c_\epsilon P_G F_\epsilon. \tag{21}$$

In particular the normalized transfer is F_ϵ on $\mathcal{H}_{\rm inv}$ and $F_\epsilon P_G$ on $\mathcal{H}$. The operator F_ϵ is positive, self-adjoint, injective and a contraction because K_ϵ has those properties and M_ϵ is positive, bounded, and invertible. Injectivity follows from $\langle f, F_\epsilon f\rangle = \langle M_\epsilon f, K_\epsilon M_\epsilon f\rangle$ and the strictly positive eigenvalues of K_ϵ. All these statements remain valid after restriction to the invariant subspace.

For several adjacent slices, the two half magnetic factors at each interior slice multiply to $e^{-\epsilon V}$. On an open time interval the two boundary slices carry half weights; on a periodic time lattice all slices have their full weight. Fubini's theorem therefore identifies the exact product of (19) with the full anisotropic Wilson integral, retaining all temporal and spatial plaquettes. In the periodic case with m time steps the exact partition identity is

$$Z^{(m)}_{\rm Wilson} = c_\epsilon^m \,\mathrm{tr}_{\mathcal{H}}(F_\epsilon^m P_G) = c_\epsilon^m \,\mathrm{tr}_{\mathcal{H}_{\rm inv}}(F_\epsilon^m). \tag{22}$$

The traces are well defined: the proof of Theorem 1 gives a trace-class single-link operator, a finite tensor product is trace class, and multiplication by bounded operators and P_G preserves that property. The integral and trace equality can also be checked by successively integrating its positive continuous compact kernels. Equation (22) is an exact finite-ϵ identity; the strong convergence below alone is not used to assert trace-norm convergence of these partition functions.

The Wilson constant and the normalization are explicitly retained as the common energy shift

$$E_{\rm scalar}(\epsilon) = -\frac{1}{\epsilon}\log c_\epsilon = \frac{N}{\epsilon}\big[2b_t - \log Z(b_t)\big]. \tag{23}$$

In particular $0 < c_\epsilon < 1$, because $e^{-2b}Z(b) = \int e^{-b(2-\mathrm{Re\,tr}\,U)}dU$ is a strictly positive average of a function at most one and strictly below one on a set of positive Haar measure. From the proved J_0 expansion,

$$e^{-2b}Z(b) = \frac{1}{2\sqrt{\pi}\,b^{3/2}}\left[1 - \frac{3}{16b} + O(b^{-2})\right].$$

Thus the shift is not finite as $\epsilon \downarrow 0$; it is a known scalar that affects all energies equally. The exact relation (21), rather than an unrecorded removal of constants, specifies which operator has the finite time limit.

5 The actual core and the strong time-continuum limit

Let $\mathscr{D}$ be the finite linear span of product matrix coefficients of all edge representations, and let $\mathscr{D}_{\rm inv} = P_G\mathscr{D}$. Peter–Weyl completeness makes $\mathscr{D}$ dense in $\mathcal{H}$ and $\mathscr{D}_{\rm inv}$ dense in $\mathcal{H}_{\rm inv}$. The gauge action mixes only matrix coefficients with the same finite list of edge spins, so $P_G\mathscr{D} \subset \mathscr{D}$. The diagonal positive operator H_0 in (13) is self-adjoint on

$$D(H_0) = \left\{ f : \sum_{\boldsymbol{j},\boldsymbol{r},\boldsymbol{s}} \left(\frac{2g_{\rm YM}^2}{a} \sum_e j_e(j_e+1) \right)^2 |f_{\boldsymbol{j},\boldsymbol{r},\boldsymbol{s}}|^2 < \infty \right\},$$

where coefficients use an orthonormal matrix-coefficient basis. Finite spectral truncation proves that $\mathscr{D}$ is an operator core: both f and $H_0 f$ are approximated by the same finite sums. The projection P_G commutes with H_0, so applying it to these graph-norm approximations proves the invariant core assertion as well.

Since V in (15) is bounded and real, the operator

$$H = H_0 + V = -\frac{2g_{\rm YM}^2}{a}\sum_e \Delta_e^T + \frac{1}{2g_{\rm YM}^2 a}\sum_{p\in \mathsf{P}_s}(2 - \mathrm{Re\,tr}\,U_p) \tag{24}$$

is self-adjoint and nonnegative on $D(H_0)$. Bounded perturbation does not change the core, since the graph norms of H_0 and H_0+V are equivalent by the two inequalities $\|Hf\| \le \|H_0 f\| + \|V\|\|f\|$ and its converse. All gauge invariance statements persist. The original-metric expression is exactly

$$H = -\frac{g_{\mathrm{YM}}^2}{2a}\sum_e \Delta_e^c + \frac{1}{2g_{\mathrm{YM}}^2 a}\sum_{p\in \mathsf{P}_s}(2 - \operatorname{Re}\operatorname{tr} U_p). \tag{25}$$

No conventional redefinition of g_{YM} was used to obtain either expression.

For $f \in \mathscr{D}$, use the exact algebraic decomposition

$$(F_\epsilon - I)f = (M_\epsilon - I)f + M_\epsilon(K_\epsilon - I)f + M_\epsilon K_\epsilon(M_\epsilon - I)f.$$

Boundedness of V gives $(M_\epsilon - I)/\epsilon \to -V/2$ in operator norm. Equation (14), the strong convergence $K_\epsilon \to I$, and the contraction bounds therefore prove

$$\frac{F_\epsilon f - f}{\epsilon} \longrightarrow -Hf \quad \text{for every } f \in \mathscr{D}, \tag{26}$$

and in particular for every $f \in \mathscr{D}_{\mathrm{inv}}$. The magnetic operator need not commute with the kinetic operator for this argument.

Theorem 3. *For every fixed $t > 0$ and every $f \in \mathcal{H}$,*

$$F_{t/m}^m f \longrightarrow e^{-tH} f \qquad (m \longrightarrow \infty). \tag{27}$$

Consequently the exact normalized Wilson transfer obeys

$$\left(c_{t/m}^{-1}\mathcal{T}_{t/m}^{\mathrm{raw}}\right)^m f = F_{t/m}^m P_G f \longrightarrow e^{-tH} P_G f. \tag{28}$$

On $\mathcal{H}_{\mathrm{inv}}$ the limit is the strongly continuous semigroup e^{-tH}, with generator (24) *or equivalently* (25)*. At $t = 0$ its identity is I on that subspace; on the unrestricted space the projected family has value P_G.*

Proof. We give the positive-contraction specialization of the product-formula argument directly; the general framework is Chernoff's theorem [2]. Put $A_\epsilon = (I - F_\epsilon)/\epsilon$. It is bounded, self-adjoint and nonnegative, and (26) gives $A_\epsilon f \to Hf$ on the operator core $\mathscr{D}$. For any $\lambda_0 > 0$ and $f \in \mathscr{D}$,

$$(\lambda_0 + A_\epsilon)^{-1}(\lambda_0 + H)f - f = (\lambda_0 + A_\epsilon)^{-1}(H - A_\epsilon)f \longrightarrow 0. \tag{29}$$

The resolvents have norms at most $1/\lambda_0$. Moreover $(\lambda_0 + H)\mathscr{D}$ is dense: for arbitrary u, apply a graph-norm approximation from $\mathscr{D}$ to $f = (\lambda_0 + H)^{-1}u$. Thus the uniform bound extends (29) to all vectors and proves $(\lambda_0 + A_\epsilon)^{-1} \to (\lambda_0 + H)^{-1}$ strongly.

For completeness this implies $e^{-tA_\epsilon} \to e^{-tH}$ strongly for each fixed $t > 0$ without any estimate of an unknown spectrum. Set $R_\epsilon = (\lambda_0 + A_\epsilon)^{-1}$ and $R = (\lambda_0 + H)^{-1}$. On $[0, 1/\lambda_0]$ define the continuous function

$$h_t(y) = \begin{cases} \exp[-t(y^{-1} - \lambda_0)], & y > 0, \\ 0, & y = 0. \end{cases}$$

Uniform polynomial approximation on this interval and strong convergence of every polynomial in the uniformly bounded R_ϵ prove $h_t(R_\epsilon) \to h_t(R)$ strongly. The spectral calculus identifies these two operators with the desired semigroups.

It remains to compare powers with this exponential. For integer $m \geq 1$ and $0 \leq x \leq 1$,

$$0 \leq e^{-mx} - (1-x)^m \leq \frac{1}{m}. \tag{30}$$

For $x \leq 1/2$, the integral identity $-\log(1-x) - x = \int_0^x u/(1-u)du \leq x^2$ gives

$$e^{-mx} - (1-x)^m \leq mx^2 e^{-mx} \leq \frac{4}{e^2 m} < \frac{1}{m}.$$

For $x \geq 1/2$ the difference is at most $e^{-m/2} \leq 2/(em) < 1/m$; the last inequality follows by maximizing $se^{-s/2}$ for $s > 0$. The endpoint $x = 1$ also satisfies the bound. Spectral calculus with $x = I - F_\epsilon = \epsilon A_\epsilon$ now yields

$$\|F_\epsilon^m - e^{-m\epsilon A_\epsilon}\| \leq 1/m.$$

Take $\epsilon = t/m$ and combine with the established strong convergence of $e^{-tA_{t/m}}$. This proves (27). Finally (21), $P_G^2 = P_G$, and commutation with F_ϵ prove (28). The same reasoning applies directly on the verified invariant core. □

6 Effective Hamiltonians and the exact finite-vacuum bridge

On $\mathcal{H}_{\text{inv}}$, positivity and injectivity allow the self-adjoint functional-calculus definition

$$H_\epsilon = -\epsilon^{-1} \log(F_\epsilon|_{\mathcal{H}_{\text{inv}}}), \qquad H_\epsilon^{\text{raw}} = E_{\text{scalar}}(\epsilon) I + H_\epsilon. \tag{31}$$

Zero is not an eigenvalue of F_ϵ, so the logarithm is densely defined even though zero is a spectral accumulation point. The operator H_ϵ is nonnegative. Equation (31) is the exact scalar-energy dictionary, and the raw and normalized effective operators have identical energy differences and eigenvectors at each ϵ.

They satisfy $H_\epsilon \to H|_{\mathcal{H}_{\text{inv}}}$ in the strong resolvent sense. Here is a direct consequence of the proof already given, which also covers arbitrary $\epsilon \downarrow 0$ rather than only t/m. Fix $t > 0$ and set $m_\epsilon = \lfloor t/\epsilon \rfloor$ and $t_\epsilon = m_\epsilon \epsilon$. The power comparison gives $\|F_\epsilon^{m_\epsilon} - e^{-t_\epsilon A_\epsilon}\| \leq 1/m_\epsilon$. For $t_\epsilon \geq t/2$, the scalar mean-value bound yields

$$\sup_{x \geq 0} |e^{-t_\epsilon x} - e^{-tx}| \leq |t - t_\epsilon| \sup_{x \geq 0} x e^{-(t/2)x} \leq \frac{2\epsilon}{et} \longrightarrow 0.$$

Thus $F_\epsilon^{m_\epsilon} \to e^{-tH}$ strongly. If $t/\epsilon = m_\epsilon + \delta$ with $0 \leq \delta < 1$, then for $0 \leq z \leq 1$,

$$0 \leq z^{m_\epsilon} - z^{m_\epsilon + \delta} \leq z^{m_\epsilon}(1-z) \leq \frac{1}{m_\epsilon + 1}.$$

The middle inequality uses $z^\delta \geq z$, and the last follows by maximizing $z^m(1-z)$. Applying spectral calculus gives $e^{-tH_\epsilon} = F_\epsilon^{t/\epsilon} \to e^{-tH}$ strongly. For any $\lambda_0 > 0$ the resolvent identity

$$(\lambda_0 + H_\epsilon)^{-1} f = \int_0^\infty e^{-\lambda_0 t} e^{-tH_\epsilon} f \, dt$$

and the uniform contraction bound permit dominated convergence in norm, proving the asserted strong resolvent convergence.

The resulting Hamiltonian is precisely the finite interacting one in the companion derivation, with the explicit substitution

$$n = 2, \qquad \kappa_e = \frac{2g_{\mathrm{YM}}^2}{a}, \qquad b_p^{\mathrm{Hamiltonian}} = \frac{1}{2g_{\mathrm{YM}}^2 a} = \frac{b_s(\epsilon)}{\epsilon}. \tag{32}$$

All magnetic interactions survive this limit. This yields a concrete state-space and operator map from the finite anisotropic Wilson integral to the vacuum-weighted finite-lattice identities; it does not identify the interacting vacuum with Haar measure.

For example, let $\psi_0 > 0$ be the normalized exact ground state of (24), E_0 its eigenvalue, and $d\mu = \psi_0^2 d\lambda$. The finite compact elliptic Hamiltonian has this unique ground state by the positivity and compact-resolvent argument in the companion paper. For any nonzero smooth gauge-invariant F with $\mu(F) = 0$, put $u = \psi_0 F$ and $D = \|u\|^2 > 0$. The exact ground-state product rule gives

$$\langle u, (H - E_0)u\rangle = \frac{2g_{\mathrm{YM}}^2}{a} \sum_{e,a} \int |X_{e,a}F|^2 d\mu. \tag{33}$$

To see the cancellation, expand $X^2(\psi_0 F) = \psi_0 X^2 F + 2(X\psi_0)XF + FX^2\psi_0$, subtract $F(H - E_0)\psi_0 = 0$, and integrate each Haar-skew derivative by parts. The remaining integral is exactly (33). Thus no component of V was omitted in passing to that form.

The strong transfer limit proves the concrete correlation identity

$$\lim_{m\to\infty} \frac{\langle u, F_{t/m}^m u\rangle}{\langle \psi_0, F_{t/m}^m \psi_0\rangle} = \langle u, e^{-t(H-E_0)}u\rangle = D \int_{[0,\infty)} e^{-t\omega} d\nu_F(\omega), \tag{34}$$

where $\nu_F(A) = D^{-1}\langle u, \mathbf{1}_A(H - E_0)u\rangle$. The denominator converges to $e^{-tE_0} > 0$. The raw scalar $c_{t/m}^m$ cancels exactly if both numerator and denominator are written using the raw transfer. The state ψ_0 in (34) is the actual limiting finite-lattice vacuum, not a constant trial state. The spectral theorem and (33) give its first moment

$$\int \omega \, d\nu_F(\omega) = \frac{(2g_{\mathrm{YM}}^2/a)\sum_{e,a} \mu(|X_{e,a}F|^2)}{\mu(|F|^2)}.$$

These are fixed-spatial-regulator statements. Strong time convergence alone does not supply a spatial continuum vacuum, a limiting spectral gap, or a gapless interacting four-dimensional quantum theory.

7 Exact checks and provenance

The companion script `check_temporal_coefficients.py` completed 237 exact scalar assertions plus exact matrix-entry checks. It verifies the two anisotropic action coefficients, Bessel recurrence coefficients through bounded formal orders, the character derivatives for dimensions 1 through 9, the resulting spin coefficients, the SU(2) metric and Casimir constants, a noncommuting temporal-plaquette example and its original gauge law, the $g = h^{-1}$ dictionary, and the magnetic endpoint weights. The all-spin and operator convergence proofs are the analytic arguments above; they are not inferred from finitely many checked spins. No numerical integration, spatial continuum calculation, or approximate vacuum was used.

References

[1] F. Peter and H. Weyl, *Die Vollständigkeit der primitiven Darstellungen einer geschlossenen kontinuierlichen Gruppe*, Mathematische Annalen **97** (1927), 737–755, doi:10.1007/BF01447892.

[2] P. R. Chernoff, *Note on product formulas for operator semigroups*, Journal of Functional Analysis **2** (1968), 238–242, doi:10.1016/0022-1236(68)90020-7. The positive self-adjoint contraction specialization needed here is proved directly in Theorem 3.